Charitap

A blockhain-based Resilient micro-donation system

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Project Committee:

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ExpoLab
Creativity Unfolded

Active Donors

100+ donors

Donations

300+ donations

Motivation

DECISION FATIGUE

Many donors struggle to choose charities wisely.

LACK OF TRANSPARENCY

Donors often have a hard time tracking where funds go.

FRICTION IN PROCESSES

Complicated donations deter people from giving regularly.

Problem Statement

People want to give, but are often held back by the pressure to make large donations in traditional donation models. On top of that, a lack of clear tracking leads to trust issues among donors who seek transparency and verifiable records of their contributions.

The Charitap Solution

Effortless Micro-Donation Integration

Charitap simplifies charitable giving by automating micro-donations through everyday online transactions, ensuring **transparency** through blockchain, and making contributions easier for users with seamless integration into their purchasing habits.

Key Features

AUTOMATED ROUND-UPS

Effortless donations from everyday purchases

FLEXIBLE PAYMENT PREFERENCES

Control over how and when to donate

BLOCKCHAIN VERIFICATION

Ensuring transparency and trust in donations

Charity Nomination

Users choose charities that they believe in

Blockchain Verification

Ensuring Trust in Donations

Every donation made through Charitap is recorded immutably on ResilientDB, providing a secure and transparent system that enhances user trust and confidence in charitable giving.

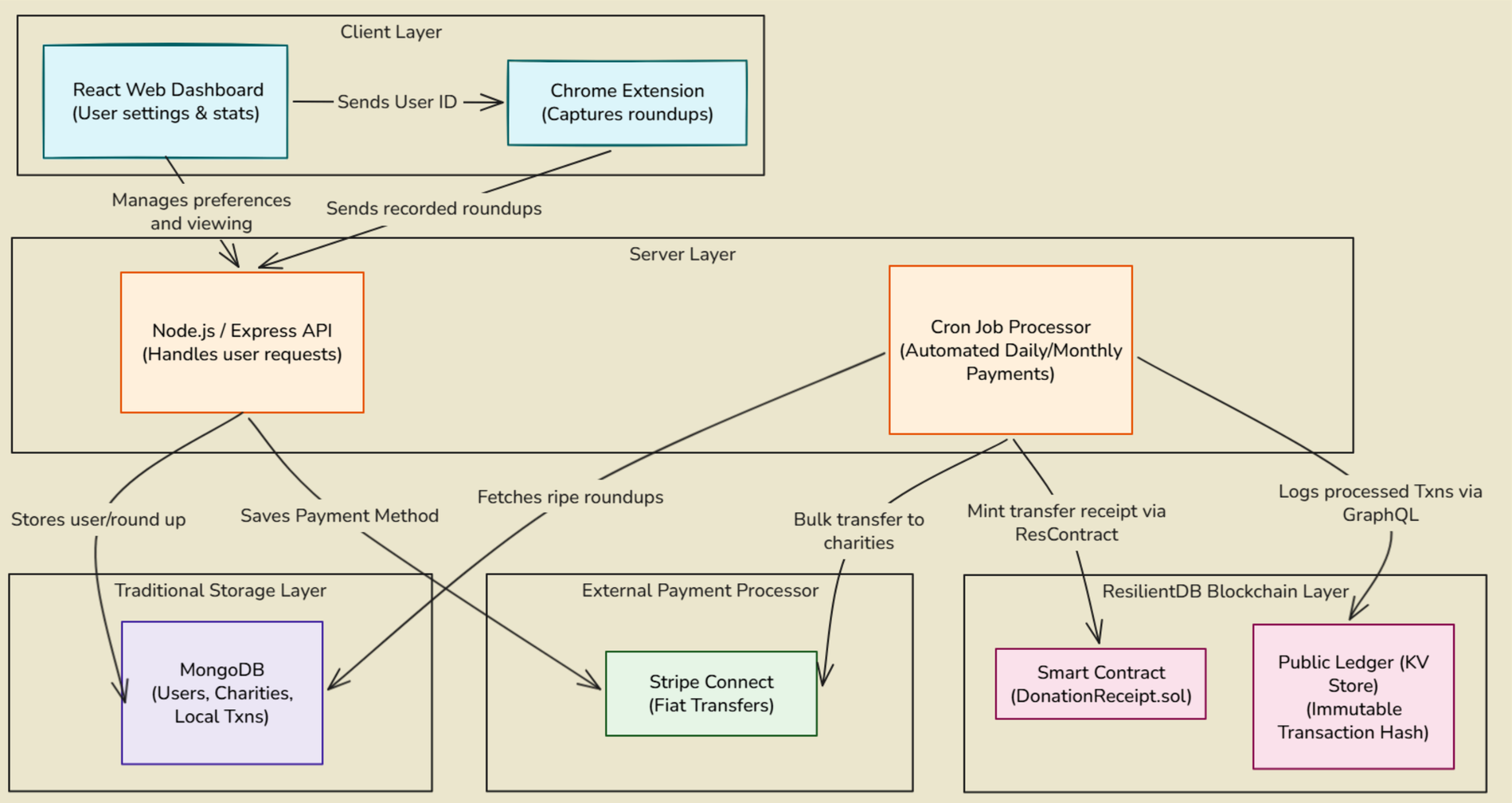
- **Public Ledger:** Permanently records the exact amount and destination of the donation so anyone can verify it.
- **Smart Contract:** Guarantees the donation receipt is mathematically accurate and charity's true total is maintained making human tampering impossible.

Apache
ResilientDB
Incubating

Tech Stack

- **Frontend:**
  - ReactJS
  - ChartJS
  - Google OAuth
  - JWT
- **Backend:**
  - NodeJS
  - ExpressJS
  - Node-Cron
  - Nodemailer
- **Database:**
  - MongoDB
- **Payments:**
  - Stripe Connect
- **Blockchain:**
  - Resilient DB Public Ledger
  - Smart Contract

Architecture



Contributions

- **Aman**:
  - Built the Chrome Extension and the roundup logic for logging the user transactions into the database.
  - Integrated Smart Contracts to mint donation receipt for user-charity transactions.
- **Dhairye:**
  - Constructed the NodeJS server, providing API endpoints for the frontend and chrome extension.
  - Setup the payment flow from the user bank account to the charity's account using Stripe Connect.
- **Himanshu:**
  - Engineered a responsive React/Tailwind dashboard with real-time Chart.js visualizations for tracking donation impact.
  - Integrated Google OAuth, Stripe, and a GraphQL-powered ResilientDB public ledger for immutable, trustless transaction transparency.

Demo